

PulmoScan AI: Deep Learning-Based Lung Cancer Detection and Classification Using CT Scan Images

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Abstract

Lung cancer is the leading cause of cancer-related mortality worldwide, with poor prognosis resulting from late-stage diagnosis. This paper presents PulmoScan AI, a four-stage deep learning framework for automated detection and classification of lung cancer subtypes from CT scan images. The proposed pipeline integrates: (1) image preprocessing including noise removal, contrast enhancement, and data augmentation; (2) multi-model classification using ResNet-50, ResNet-101, GoogLeNet (InceptionV3), and EfficientNet-B0; (3) lung nodule segmentation using U-Net; and (4) prediction explainability through Grad-CAM heatmaps. Evaluated on the IQ-OTH/NCCD Kaggle dataset (1575 CT images, four classes), GoogLeNet achieved the highest testing accuracy of 91.75% with a Mean AUC of 0.988. The system is deployed as a Flask web application to support clinical decision-making.

Keywords: Lung Cancer, CT Scan, Deep Learning, ResNet, GoogLeNet, InceptionV3, EfficientNet, U-Net, Grad-CAM, CNN Classification, Medical Image Processing, PulmoScan AI

1. Introduction

Lung cancer accounts for over 1.8 million deaths annually worldwide [1]. More than 85% of cases are diagnosed at advanced stages (Stage III or IV), where five-year survival rates drop below 10%. Early detection using non-invasive imaging—particularly Computed Tomography (CT) scanning—is therefore critical for improving patient outcomes.

CT scans provide high-resolution cross-sectional images of lung tissue and are the gold standard for lung cancer screening. However, manual radiological interpretation is time-consuming, subjective, and requires expert knowledge. Automated deep learning systems can address these limitations by providing fast, consistent, and scalable analysis of CT scan images.

This paper presents PulmoScan AI—a four-stage deep learning pipeline that combines preprocessing, classification with multiple CNN architectures, U-Net segmentation, and Grad-CAM explainability. Our system classifies CT images into four lung tissue categories: Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal.

2. Related Work

Significant research has been conducted on deep learning-based lung cancer detection. Ardila et al. [1] proposed an end-to-end 3D CNN trained on NLST low-dose CT data achieving high lung cancer prediction accuracy. Setio et al. [2] applied multi-view CNNs for pulmonary nodule detection on the LUNA16 dataset. Shen et al. [3] demonstrated multi-scale CNN effectiveness on LIDC-IDRI for nodule detection.

He et al. [4] introduced ResNet with skip connections that solved the vanishing gradient problem, enabling training of networks exceeding 100 layers. Szegedy et al. [5] proposed GoogLeNet featuring Inception modules for multi-scale parallel convolution. Tan and Le [6] developed EfficientNet using compound scaling of depth, width, and resolution.

For segmentation, Ronneberger et al. [7] introduced U-Net—an encoder-decoder architecture with skip connections designed for biomedical image segmentation with limited training data. Selvaraju et al. [8] proposed Grad-CAM, providing class-discriminative visual explanations for CNN decisions, critical for clinical interpretability. Zhao et al. [9] demonstrated CNN-based lung cancer detection achieving high accuracy on medical CT datasets.

3. Dataset Description

The IQ-OTH/NCCD Lung Cancer Dataset sourced from Kaggle was used for all experiments. The dataset comprises CT scan images categorized into four classes representing different lung tissue conditions. All images were resized to 224×224 pixels before training to match CNN input requirements. An 80:20 train-validation split was applied, yielding 1260 training and 315 validation images.

Table 1: IQ-OTH/NCCD Dataset Distribution

Class	Description	Total Images	Train (80%)	Validation (20%)
Adenocarcinoma	Most common; outer lung regions	~500	~400	~100
Large Cell Carcinoma	Fast-growing; any lung location	~300	~240	~60
Squamous Cell Carcinoma	Central airways; linked to smoking	~400	~320	~80
Normal	Healthy lung tissue	~375	~300	~75
Total	—	1575	1260	315

4. Proposed Methodology

PulmoScan AI is a structured four-stage deep learning pipeline for lung cancer detection. Figure 1 illustrates the complete implementation workflow from data acquisition through explainability and final classification output.

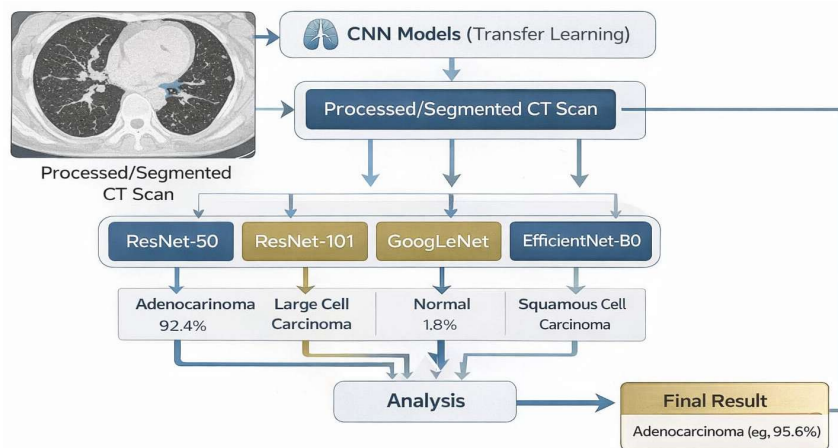


Fig. 1: PulmoScan AI Implementation Pipeline — Data Acquisition → Preprocessing → Model Training → Evaluation → Grad-CAM Explainability → Detection

4.1 Stage 1: Image Preprocessing

Raw CT scan images were subjected to the following preprocessing steps before being passed to the classification and segmentation models:

- **Noise Removal:** Gaussian filtering applied to reduce CT acquisition noise and smooth image gradients.
- **Contrast Enhancement:** Histogram equalization applied to improve visibility of pulmonary nodules and tissue boundaries.
- **Resizing:** All images uniformly resized to 224×224 pixels (299×299 for GoogLeNet) to match CNN input dimensions.
- **Data Augmentation:** Random horizontal flipping, rotation ($\pm 15^\circ$), and zoom ($0.9\text{--}1.1\times$) applied to training images to reduce overfitting.
- **Normalization:** Pixel intensities normalized to $[0, 1]$ range for stable gradient computation during backpropagation.

4.2 Stage 2: Multi-Model CNN Classification

Four CNN architectures were implemented using transfer learning with pre-trained ImageNet weights. Each model's final classification layer was replaced with a 4-class Softmax output layer and fine-tuned on the lung CT dataset.

ResNet-50 & ResNet-101 (Residual Networks): Residual Networks address the vanishing gradient problem through skip connections that allow direct gradient flow across layers. Each residual block computes $H(x) = F(x) + x$, where $F(x)$ is the residual function learned by 2–3 convolutional layers and x is the identity shortcut connection. ResNet-50 uses 50 layers organized into 4 stages (3×, 4×, 6×, 3× bottleneck blocks), while ResNet-101 extends to 101 layers with a deeper Stage 3 (23× blocks) for richer feature extraction. Both architectures terminate with Global Average Pooling followed by a Softmax classifier.

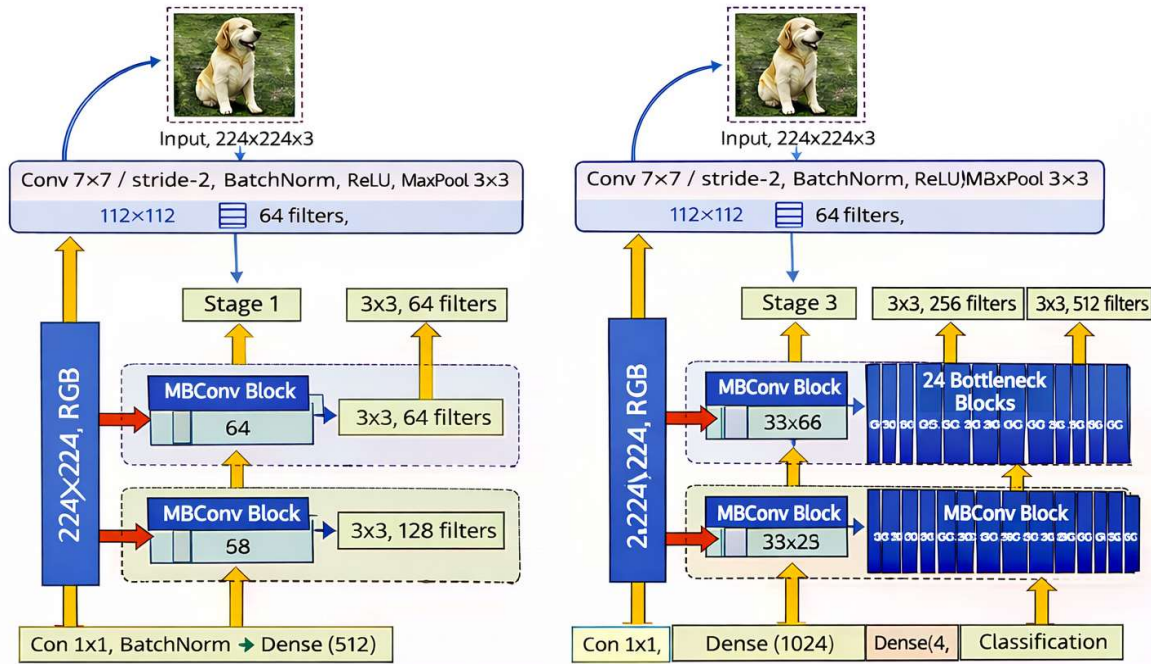


Fig. 2: ResNet Architecture — Comparison of ResNet-50, ResNet-101 with residual block detail showing skip connections

GoogLeNet (InceptionV3): GoogLeNet employs Inception modules that apply parallel convolution filters of sizes 1×1 , 3×3 , and 5×5 on the same feature map, then concatenate their outputs. This multi-scale architecture captures features at different spatial resolutions simultaneously. The InceptionV3 variant processes 299×299 input through a stem layer, three Inception-A modules, a Reduction-A block, four Inception-B modules, a Reduction-B block, two Inception-C modules, global average pooling, and final classification. Factorized convolutions (replacing $n \times n$ with $1 \times n + n \times 1$) reduce computational cost while auxiliary classifiers improve gradient propagation.

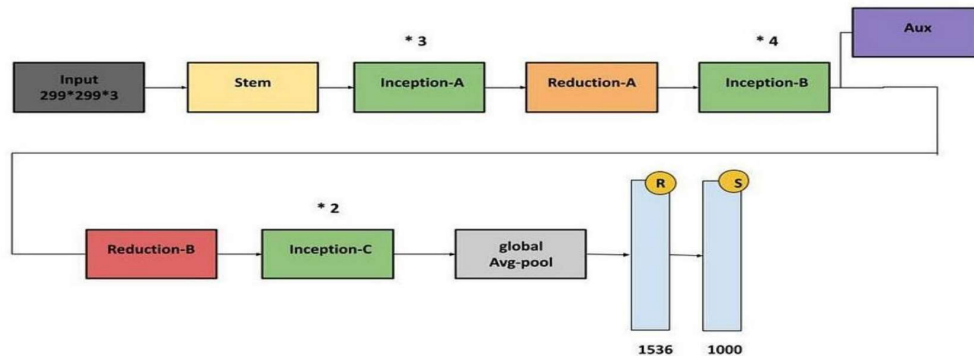


Fig. 3: GoogLeNet (InceptionV3) Architecture — Input 299×299 through Stem, Inception-A/B/C modules, Reduction blocks, to Softmax classification

EfficientNet-B0 (Mobile Inverted Bottleneck CNN): EfficientNet applies compound scaling to simultaneously increase network depth, width, and input resolution using a fixed scaling coefficient. EfficientNet-B0 is the baseline variant comprising 237 layers built from MBConv (Mobile Inverted Bottleneck Convolution) blocks with depthwise separable convolutions and Swish activation. The architecture scales from 16 filters in the first MBConv1 block to 320 filters in the final MBConv6 block, with integrated Squeeze-and-Excitation attention within each block. Despite its 237-layer depth, it is the most computationally lightweight model evaluated.

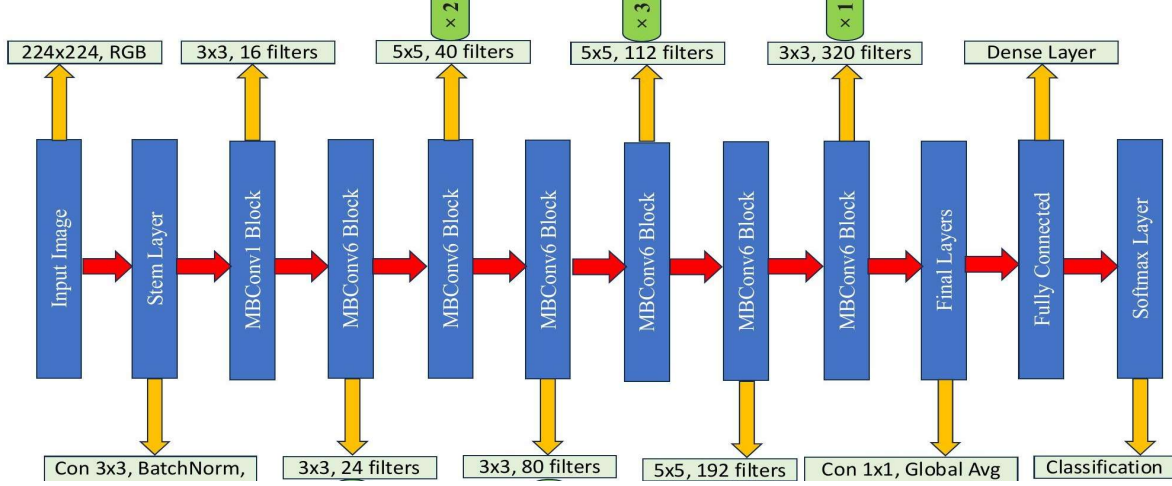


Fig. 4: EfficientNet-B0 Architecture — MBConv block stack from Stem (224×224) through 7 MBConv stages to Global Average Pooling and Softmax

Table 2: CNN Architecture Comparison Summary

Model	No. of Layers	Input Size	Key Innovation	Activation
ResNet-50	50	224×224	Skip connections ($H(x) = F(x)+x$)	ReLU
ResNet-101	101	224×224	Deeper residual blocks (23× Stage 3)	ReLU
GoogLeNet (InceptionV3)	22	299×299	Inception modules (multi-scale parallel conv)	ReLU
EfficientNet-B0	237	224×224	MBConv + compound scaling + SE blocks	Swish

4.3 Stage 3: Lung Region Segmentation (U-Net)

U-Net is a fully convolutional encoder-decoder network designed for biomedical image segmentation. The architecture consists of a contracting encoder path that progressively downsamples feature maps using max-pooling (capturing semantic context), and a symmetric expanding decoder path that upsamples using transposed convolutions (restoring spatial resolution).

The defining feature of U-Net is its skip connections that concatenate encoder feature maps with corresponding decoder layers. This preserves fine spatial details lost during downsampling and enables precise boundary delineation. The encoder extracts feature maps at 4 scales (64→128→256→512 channels), with a bottleneck at 1024 channels, before the decoder reconstructs the segmentation map.

In PulmoScan AI, U-Net segments the lung region from CT scan images, generating binary masks that isolate pulmonary tissue from surrounding anatomical structures (ribs, heart, mediastinum). This segmentation preprocessing step focuses the classification model's attention on relevant lung tissue.

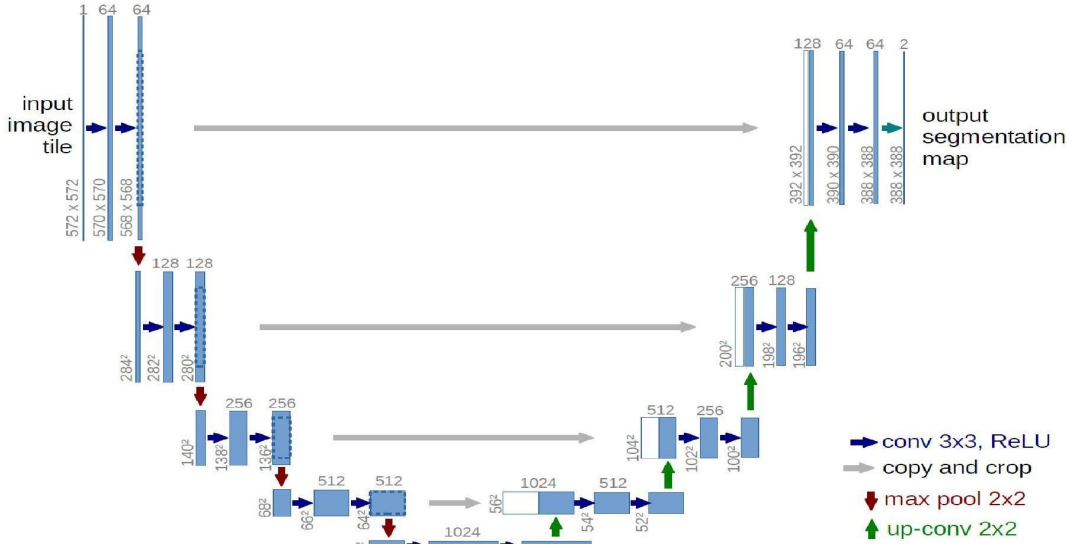


Fig. 5: U-Net Architecture — Encoder (contracting path) with max-pooling, bottleneck at 1024 channels, Decoder (expanding path) with up-convolution and skip connections

4.4 Stage 4: Explainability via Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) produces visual explanations for CNN classification decisions without requiring architectural modifications. For a predicted class c , Grad-CAM computes the gradient of the class score y^c with respect to the feature maps A^k of the final convolutional layer. These gradients are globally average-pooled to obtain channel importance weights $\alpha^{kc} = (1/Z) \sum_i \sum_j (\partial y^c / \partial A_{ij}^k)$. The Grad-CAM heatmap $L^c = \text{ReLU}(\sum_k \alpha^{kc} A^k)$ highlights regions positively influencing the prediction.

Heatmaps are upsampled to input image resolution and superimposed on the original CT scan. Red/yellow regions indicate high discriminative importance; blue regions indicate low contribution. In PulmoScan AI, Grad-CAM enables radiologists to verify that model decisions are anatomically consistent, enhancing clinical trust.

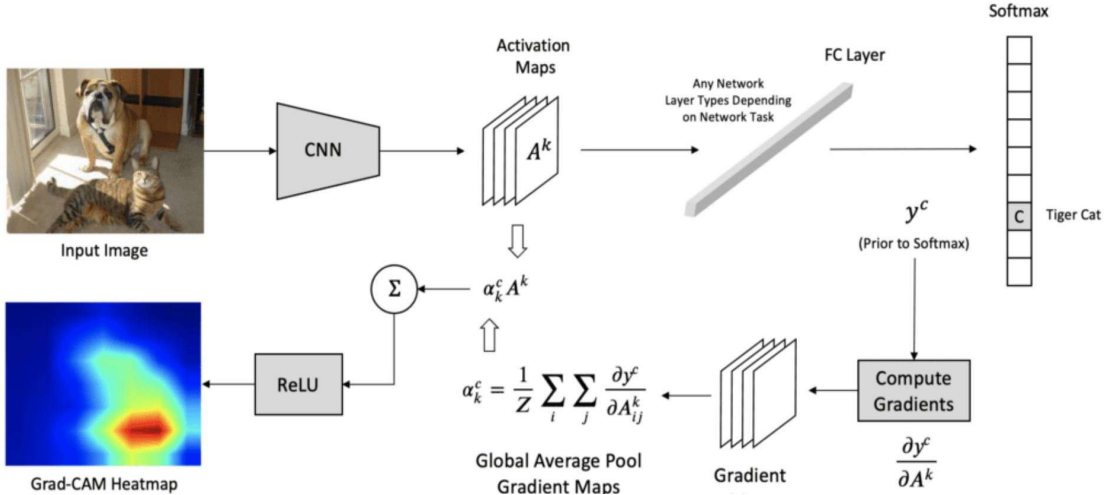


Fig. 6: Grad-CAM Architecture — Gradient flow from class score through activation maps, global average pooling for channel weights, and heatmap generation via ReLU

5. Existing System vs. Proposed System

Traditional lung cancer detection systems relied on handcrafted feature extraction techniques such as texture analysis, morphological pattern recognition, and Support Vector Machine (SVM) classifiers. These methods require domain-expert manual intervention, are time-consuming, and lack visual explainability.

Table 3: Comparative Analysis — Existing System vs. PulmoScan AI

Parameter	Existing System (SVM-Based)	Proposed PulmoScan AI
Feature Extraction	Manual (handcrafted)	Automatic (CNN-learned)
Classification	SVM / shallow ML	ResNet, GoogLeNet, EfficientNet
Segmentation	Not available	U-Net lung region masking
Explainability	No visual output	Grad-CAM heatmaps
Best Accuracy	$\leq 80\%$	91.75% (GoogLeNet)
Human Effort	High (expert dependent)	Low (automated pipeline)
Deployment	Offline / research only	Flask web application
Scalability	Limited	Cloud-ready deployment

6. Experimental Results and Discussion

6.1 Classification Performance

All four CNN models were trained for 25 epochs using Adam optimizer with learning rate 0.001 and categorical cross-entropy loss. Performance was evaluated using testing accuracy, precision, recall, and ROC-AUC. Table 4 summarizes overall model performance.

Table 4: Classification Model Performance Comparison

Model	Train Acc. (%)	Test Acc. (%)	Precision (%)	Recall (%)	Mean AUC
ResNet-50	91.2	90.48	90.5	90.1	0.984
ResNet-101	92.0	90.16	90.2	90.0	0.988
GoogLeNet (InceptionV3)	93.5	91.75	91.8	91.5	0.988
EfficientNet-B0	75.0	71.75	72.0	71.5	0.941

GoogLeNet achieved the highest testing accuracy of 91.75% with the best precision (91.8%) and recall (91.5%). ResNet-101 achieved a marginally higher Mean AUC (0.988) matching GoogLeNet, confirming strong discriminative capability. EfficientNet-B0's lower performance (71.75%) is attributed to its lightweight baseline design requiring larger datasets for optimal fine-tuning.

6.2 ResNet-50 Results

ResNet-50 achieved 90.48% overall testing accuracy. Per-class accuracy was: Adenocarcinoma 95.8%, Large Cell Carcinoma 96.1%, Normal 98.1%, and Squamous Cell Carcinoma 75.6%. The ROC analysis yielded Mean AUC = 0.984, with the Normal class achieving AUC = 0.992.



Fig. 7: ResNet-50 — Per-Class Accuracy Breakdown (left) and Confusion Matrix + ROC Curve (right) | Overall Accuracy: 90.48%, Mean AUC: 0.984

6.3 ResNet-101 Results

ResNet-101 achieved 90.16% overall testing accuracy. Per-class accuracy: Adenocarcinoma 94.2%, Large Cell Carcinoma 96.1%, Normal 96.3%, Squamous Cell Carcinoma 77.8%. The deeper architecture achieved Mean AUC = 0.988, with Normal class achieving perfect AUC = 1.000, demonstrating excellent discriminability for healthy lung tissue.



Fig. 8: ResNet-101 — Per-Class Accuracy Breakdown (left) and Confusion Matrix + ROC Curve (right) | Overall Accuracy: 90.16%, Mean AUC: 0.988

6.4 GoogLeNet (InceptionV3) Results — Best Performing Model

GoogLeNet achieved the highest overall testing accuracy of 91.75%, outperforming all other models. Per-class accuracy: Adenocarcinoma 95.8%, Large Cell Carcinoma 96.1%, Normal 98.1%, Squamous Cell Carcinoma 80.0%. Mean AUC = 0.988 with Normal class AUC = 1.000. The Inception module's multi-scale feature extraction capability is credited for superior performance on this CT classification task.

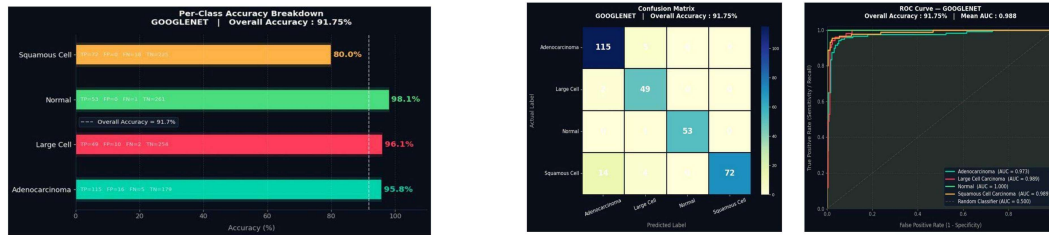


Fig. 9: GoogLeNet — Per-Class Accuracy Breakdown (left) and Confusion Matrix + ROC Curve (right) | Overall Accuracy: 91.75%, Mean AUC: 0.988 ★ Best Model

6.5 EfficientNet-B0 Results

EfficientNet-B0 achieved the lowest overall accuracy of 71.75%. Notable weakness observed in Adenocarcinoma classification (51.7%), with many instances misclassified as Large Cell Carcinoma (31 cases) and Normal (12 cases). This is consistent with EfficientNet-B0 being the smallest EfficientNet variant. Mean AUC = 0.941, with Adenocarcinoma AUC of only 0.894 confirming classification difficulty.

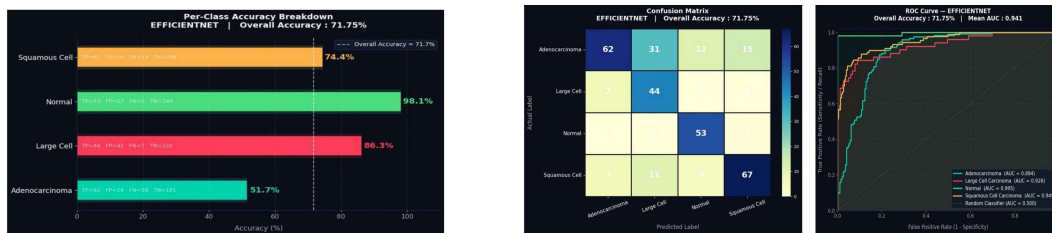


Fig. 10: EfficientNet-B0 — Per-Class Accuracy Breakdown (left) and Confusion Matrix + ROC Curve (right) | Overall Accuracy: 71.75%, Mean AUC: 0.941

6.6 ROC-AUC Analysis

Table 5 provides a comprehensive per-class AUC comparison across all models. The Normal class consistently achieves the highest AUC across all models due to its distinctly different CT appearance (clear lung fields, no nodules). Squamous Cell Carcinoma consistently shows the lowest per-class accuracy due to visual overlap with Adenocarcinoma patterns.

Table 5: Per-Class ROC-AUC Comparison Across All Models

Model	Adenocarcinoma	Large Cell	Normal	Squamous Cell	Mean AUC
ResNet-50	0.966	0.986	0.992	0.992	0.984
ResNet-101	0.977	0.993	1.000	0.981	0.988
GoogLeNet	0.973	0.989	1.000	0.989	0.988
EfficientNet-B0	0.894	0.926	0.995	0.949	0.941

6.7 U-Net Segmentation Results

The U-Net segmentation model was trained over 20 epochs with binary cross-entropy loss and Adam optimizer. Training loss decreased from 0.48 at epoch 0 to approximately 0.05 at epoch 20, demonstrating strong model convergence. Figure 11 shows the training loss curve alongside input CT images, ground truth masks, and predicted segmentation masks.

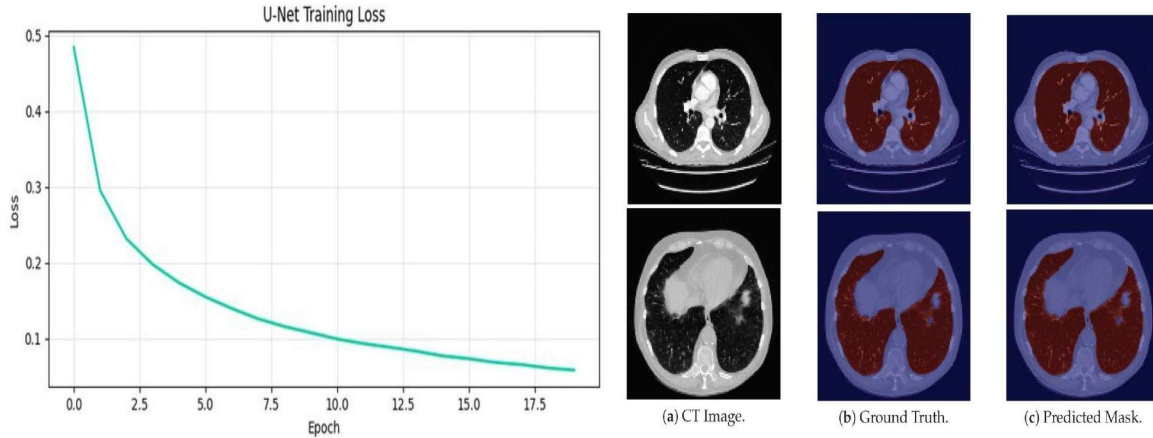


Fig. 11: U-Net Segmentation Results — Training loss convergence curve (left); CT image, ground truth mask, and predicted mask comparisons (right)

Predicted masks closely correspond with ground truth lung region annotations, confirming effective isolation of pulmonary tissue from surrounding structures including the heart, ribs, and mediastinum. The segmentation provides a focused region of interest for the downstream classification models.

7. Discussion

The experimental results demonstrate that GoogLeNet (InceptionV3) is the most suitable architecture for lung cancer subtype classification from CT scan images among the evaluated models. The multi-scale Inception modules enable simultaneous extraction of fine-grained nodule texture features and coarse-grained lung structural patterns, which is particularly valuable for distinguishing between visually similar cancer subtypes.

The consistent weakness observed in Squamous Cell Carcinoma classification across all models (75.6%–80.0%) reflects the inherent visual similarity between squamous cell patterns and adenocarcinoma in CT imaging. Future work incorporating 3D volumetric analysis or multi-view fusion may address this limitation.

The U-Net segmentation stage contributes to interpretability by isolating the lung region before classification, reducing the risk of the CNN focusing on non-lung anatomical structures. The Grad-CAM explainability layer is critical for clinical deployment, enabling radiologists to verify anatomical consistency of model decisions—a key requirement for regulatory acceptance of AI-assisted diagnosis tools.

EfficientNet-B0's underperformance (71.75%) highlights that compound scaling benefits are most pronounced in larger variants (B4–B7) or with larger training datasets. Fine-tuning strategies or ensemble approaches may improve its performance on medical imaging tasks.

8. Conclusion

This paper presented PulmoScan AI, a comprehensive four-stage deep learning framework for lung cancer detection and classification from CT scan images. The system integrates image preprocessing, multi-model CNN classification (ResNet-50, ResNet-101, GoogLeNet, EfficientNet-B0), U-Net lung segmentation, and Grad-CAM visual explainability into a unified pipeline.

Experimental evaluation on the IQ-OTH/NCCD dataset demonstrated that GoogLeNet achieved the best classification performance with 91.75% testing accuracy and Mean AUC of 0.988. U-Net effectively segmented lung regions with training loss converging to 0.05 over 20 epochs. Grad-CAM heatmaps confirmed anatomically consistent model focus, particularly highlighting peripheral regions for Adenocarcinoma and central airway regions for Squamous Cell Carcinoma—consistent with known cancer growth patterns.

PulmoScan AI reduces reliance on manual feature engineering, eliminates diagnostic subjectivity, and provides an explainable AI-driven tool suitable for integration into clinical screening workflows via its Flask web application deployment.

9. Future Scope

- Extension to 3D volumetric CT analysis using 3D-CNN and Vision Transformer (ViT) architectures for improved spatial feature modeling.
- Integration of multimodal imaging (CT + PET fusion) for enhanced diagnostic accuracy and metabolic activity assessment.
- Deployment in real-time hospital information systems with full DICOM image format compatibility.
- Use of larger multi-institutional datasets and federated learning for improved model generalizability across diverse patient populations.
- Exploration of EfficientNet-B4/B7 and ConvNeXt architectures for potential accuracy improvements.
- Integration of automatic nodule measurement and staging classification to extend system utility for oncologists.

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